Polynomial Regression: Geothermal Plant Modelling

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1. Analysis:

Each problem has 1000 training samples and 1000 test samples.

One thing I noticed was that a high-degree polynomial creates a large number of features. With six variables, even degree 10 gives a very large feature space. I decided to compare different polynomial degrees along with Ridge and Lasso regularisation instead of directly using the maximum allowed degree.

For var1, the test inputs contain more values close to the limits of the input range than the training data. A sizeable number of values are exactly 1/-1. This means that some test points are farther from the centre of the training data, the model also needs to behave reasonably near the boundaries.

For var2, the train and test distributions looked much more similar. The fraction of clipped values near 1 & -1 was also similar in the two datasets.

2. Method:

I used polynomialfeatures from scikit-learn to generate polynomial terms up to a chosen degree.

I did not want to directly choose the maximum allowed degree because that could introduce a very large number of terms and lead to overfitting.

So I compared different polynomial degrees and regularisation strengths using cross-validations.

I considered both Ridge and Lasso regression:

Ridge regression adds L2 and mainly shrinks the coefficients towards zero but doesnt eliminate any coefficient. Its useful for cases where many polynomial terms may contribute to the prediction.

Lasso regression adds L1 and can make many coefficients exactly zero. This looked useful for var1, where the higher-dimensional polynomial expansion can become difficult to control.

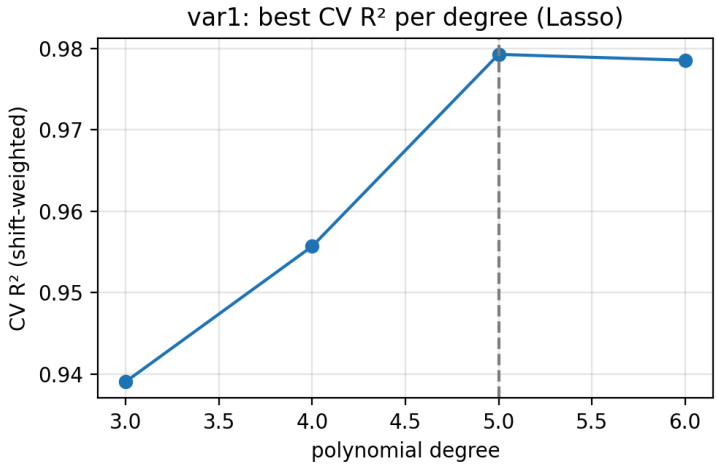
For var1, I also looked at the difference between the training and test input distributions. Since the test points were more concentrated near the boundaries, I used importance weighting during validation so that training samples more similar to the test distribution had more influence on the validation error.

After that, I selected one degree and L1/L2 values for each problem and then trained the final model using all training samples.

3. var1 : Degree & Model

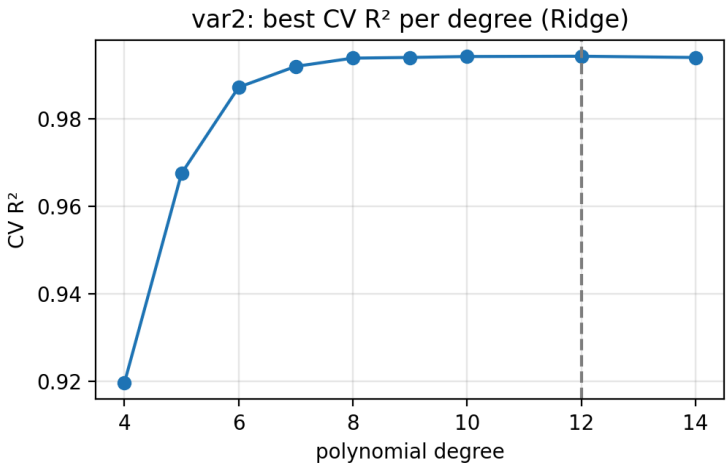
Degree	alpha(best)	CV-R2

3	0.001	0.9390
4	0.005	0.9557
5 (chosen)	0.003	0.9793
6	0.003	0.9786



4. var2 : Degree & Model

Degree	alpha (best)	CV R2
4	~0	0.9196
5	0.01	0.9676
6	0.01	0.9873
7	0.01	0.9921
8	0.01	0.9940
9	0.01	0.9941
10	0.1	0.9943
12 (chosen)	0.1	0.9944
14	0.1	0.9941



5. Results Summary:

Var 1 – LASSO – Degree 5 – alpha 0.003 – CV R2 0.9669 – Train R2 0.9764

Var 2 – RIDGE – Degree 12 – alpha 0.1 – CV R2 0.9944 – Train R2 0.9963

Repository link: